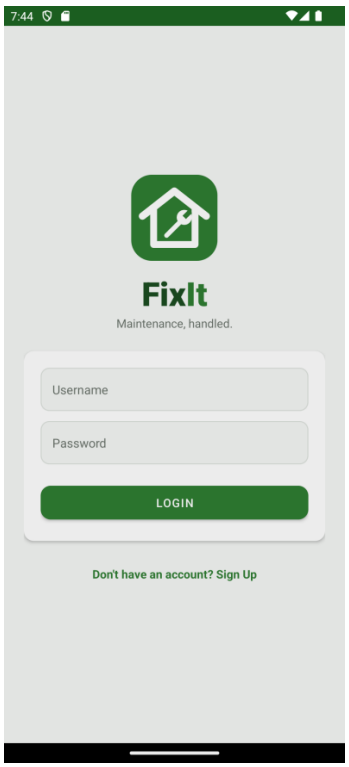
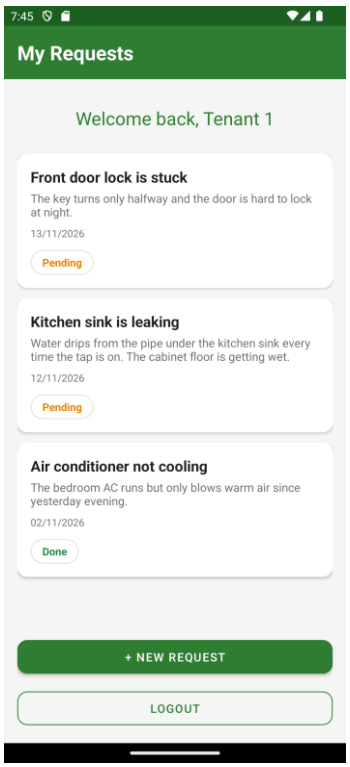
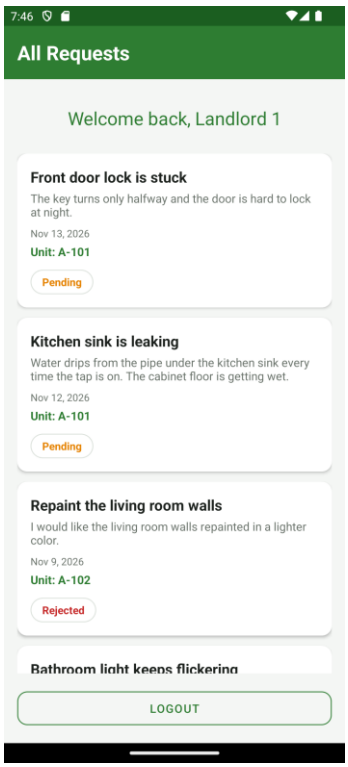
FixIt: Project Documentation

Landlord–Tenant Maintenance App | Android (Java, XML, Room) | Version 1.0 | 14 November 2026







1. Overview

FixIt connects tenants and landlords around maintenance. A tenant reports a problem in their unit; the landlord sees the request, then completes it or rejects it with a reason. Both sides see the current status.

The app runs entirely on the device: accounts and requests are stored in a local SQLite database through Room. There is no server.

2. Features

Tenants

- Sign up with name, username, email, password, unit number and the username of their landlord.
- Submit maintenance requests (title and description).
- See their own requests with a coloured status badge: Pending (orange), Done (green) or Rejected (red).
- Edit or delete a request while it is Pending.
- Tap a Rejected request to see the reasons the landlord gave.

Landlords

- See only the requests from their own tenants, Pending first, with each tenant's unit number.
- Open a request to read the details and the tenant's name, unit and email (tap the + button).

- Mark a request Done, or Reject it by ticking reasons from a checklist. A request can be reopened.
- Contact the tenant by email, with the request title as the subject.
- Receive a notification when a new request is submitted.

Everyone

- Welcome message with the user's name on the home screens.
- Password rule at sign up: at least 4 letters and 3 numbers.
- One consistent design: green title bars, white cards, rounded buttons and Back buttons on inner screens.

3. Demo accounts and data

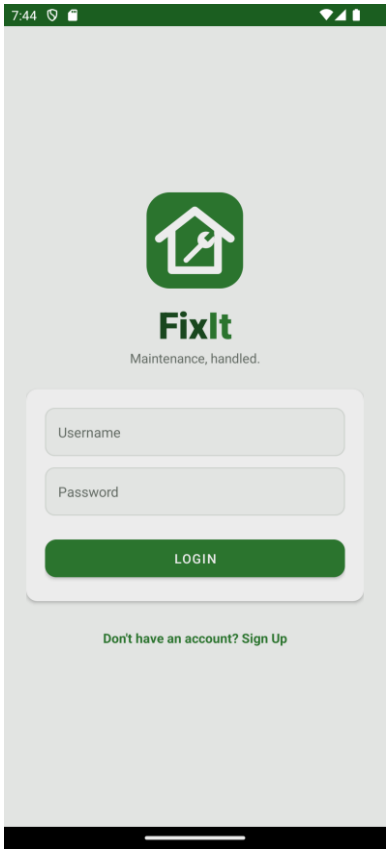
The app creates these accounts and eight sample requests the first time it starts with an empty database.

Role	Username	Password	Unit	Landlord
Landlord	`landlord1`	`landlord123`		
Landlord	`landlord2`	`landlord123`		
Tenant	`tenant1`	`tenant123`	A-101	landlord1
Tenant	`tenant2`	`tenant123`	A-102	landlord1
Tenant	`tenant3`	`tenant123`	B-201	landlord2

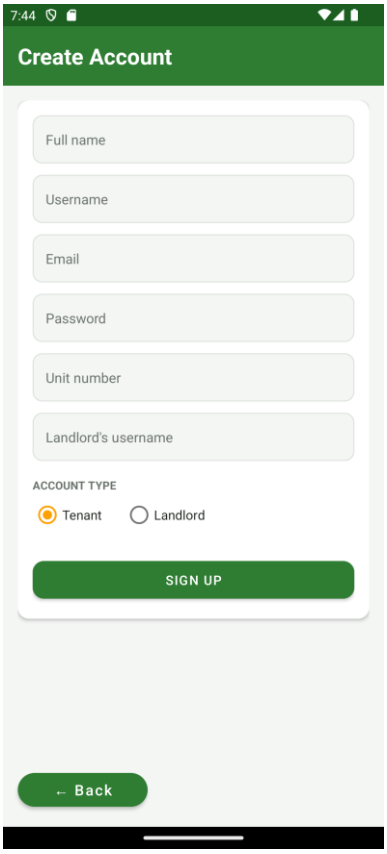
Requests: Tenant 1 has three (two Pending, one Done), Tenant 2 has two (one Done, one Rejected), Tenant 3 has three (two Pending, one Rejected).

4. Screens

4.1 Login and Sign Up



Login

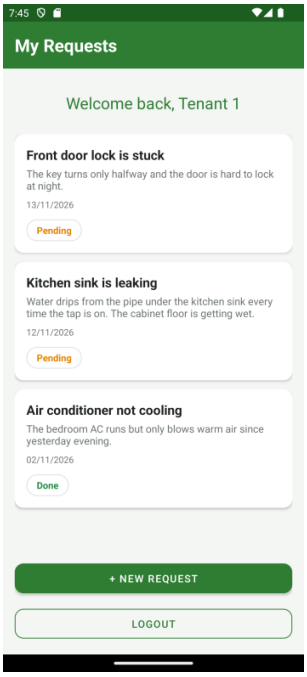


Sign Up

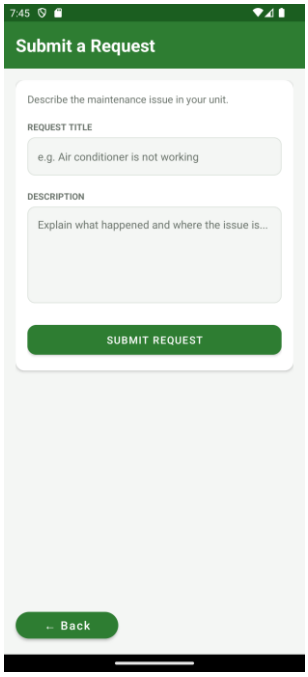
Login checks the username and password. Empty fields show “Please fill in all fields”; wrong credentials show “Username or password is incorrect”. On success the user id and role are saved and the tenant or landlord home screen opens.

Sign Up asks for name, username, email, password, account type, and (for tenants) unit number and the landlord's username. The unit and landlord fields are hidden for landlords.

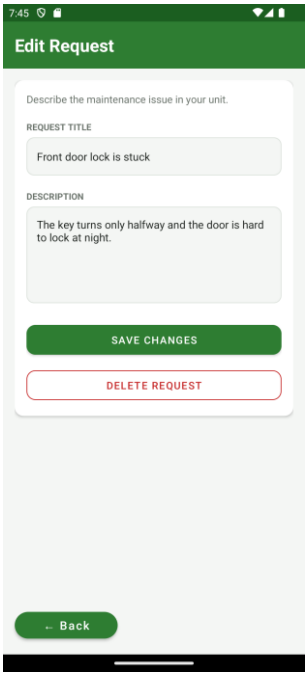
4.2 Tenant screens



My Requests

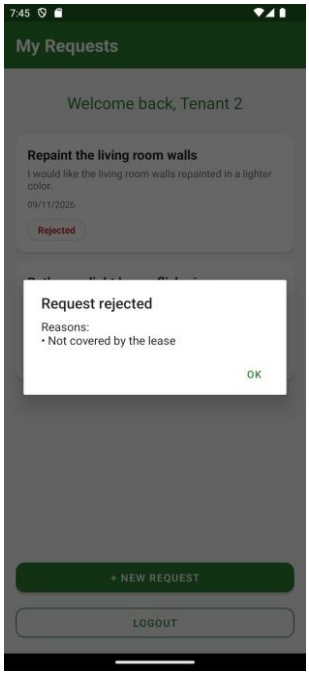


New request



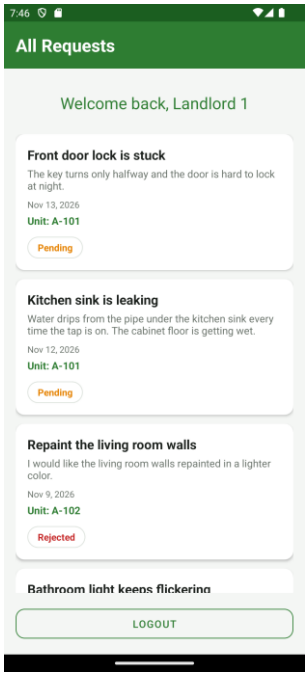
Edit request

My Requests lists the tenant's own requests; the whole page scrolls, and “+ New Request” and Logout stay at the bottom. Tapping a Pending request opens the same form as New Request, filled in, with Save Changes and Delete Request (with a confirmation).

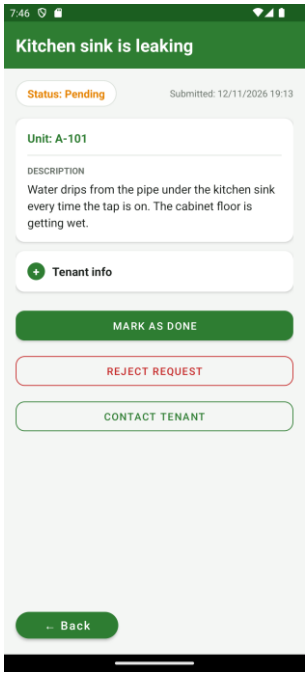


Rejected request: reasons

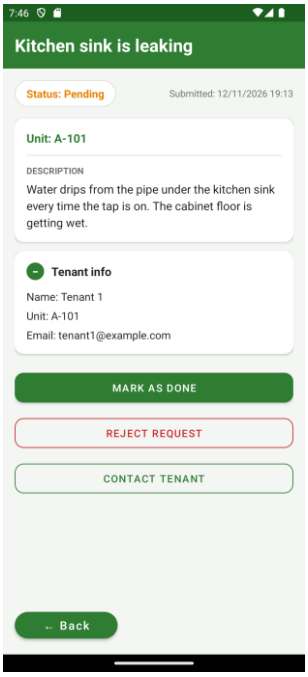
4.3 Landlord screens



All Requests

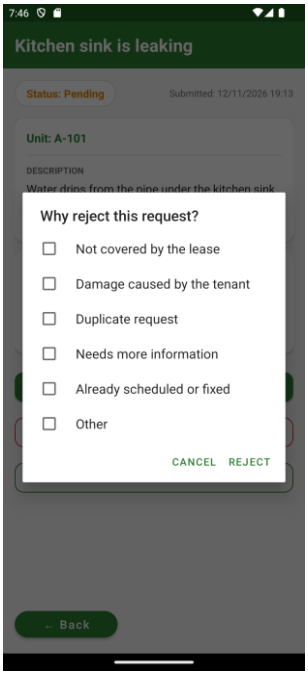


Request detail

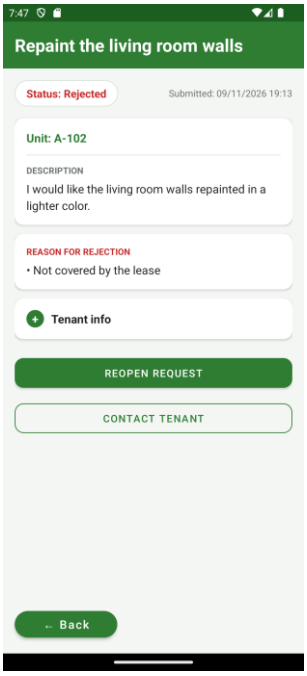


Tenant info (+)

All Requests shows the requests of the landlord's tenants, Pending first. Request Detail shows the status badge, date, unit and description. The + button opens the tenant's name, unit and email.



Reject: reasons checklist



A rejected request

Reject Request opens a checklist of reasons (at least one must be ticked). The request then shows as Rejected with the reasons; Reopen Request puts it back to Pending.

5. How a request moves

1. A tenant submits a request. It is saved as Pending and a NEW_REQUEST broadcast is sent.
2. The receiver shows a notification and tells the landlord screen to refresh.

3. The landlord opens the request and either marks it Done or rejects it with reasons.
4. The tenant sees the new status badge, and the reasons for a rejection.
5. The landlord can reopen a Done or Rejected request, which clears any reasons and sets it back to Pending.

Who can do what

Status	Tenant	Landlord
Pending	Edit, delete	Mark Done, Reject, contact tenant
Done	View only	Mark as Pending, contact tenant
Rejected	Tap to read the reasons	Reopen Request, contact tenant

6. Data model

Table users

Column	Type	Notes
user_id	INTEGER	Primary key, auto-generated
name	TEXT	Shown in the welcome message and tenant info
username	TEXT	Unique (index)
password	TEXT	Stored as plain text
email	TEXT	Used by Contact Tenant
role	TEXT	“tenant” or “landlord”
unit_number	TEXT	Tenants only
landlord_id	INTEGER	Tenants only: the user_id of their landlord

Table requests

Column	Type	Notes
request_id	INTEGER	Primary key, auto-generated
tenant_id	INTEGER	The tenant who submitted it (users.user_id)
title	TEXT	Short summary
description	TEXT	Details
status	TEXT	Pending, Done or Rejected
created_at	INTEGER	Milliseconds since 1970
reject_reasons	TEXT	Reasons joined with “|”; only set while Rejected

A landlord's requests are found through the link tenant → landlord: requests whose tenant has landlord_id equal to the landlord's user_id.

Database versions

Version	Change
1	Users and requests tables
2	Added users.landlord_id and the unique username index (old data was rebuilt)

Version	Change
3	Added requests.reject_reasons, with a migration that keeps existing data

7. Architecture

Class	Responsibility
LoginActivity	Sign in, save the session, seed the demo data on an empty database
SignupActivity	Create accounts, validate fields, password rule and landlord username
TenantHomeActivity, RequestAdapter	Tenant's request list; tap to edit or view rejection reasons
NewRequestActivity	Create a request, or edit/delete a Pending one
LandlordHomeActivity	Landlord's request list, notification permission, refresh receiver
RequestDetailActivity	Details, tenant info, Done/Pending, Reject with reasons, contact tenant
NewRequestReceiver	Receives NEW_REQUEST, shows the notification, sends a refresh broadcast
User, UserDao, Request, RequestDao	Room entities and queries
AppDatabase	Room database fixit_db (version 3), singleton, migration 2→3

Key design decisions

- Database work runs on a background thread (an executor), never on the main thread; results are posted back to the UI.
- The logged-in user is kept in SharedPreferences file “fixit_prefs” with keys user_id and role.
- Screens share one look through styles in themes.xml and colours in colors.xml.
- Notifications use a broadcast restricted to the app's own package, so other apps cannot trigger them.

8. Business rules

Rule	Where it is enforced
Password has at least 4 letters and 3 numbers	Sign Up (toast on failure). Login does not re-check it.
Username is unique	Sign Up check and a unique database index
Email contains “@”	Sign Up
Tenants need a unit number and an existing landlord username	Sign Up (“Landlord not found”)
Landlords see only their own tenants' requests	RequestDao.getForLandlord
Only the owner can edit or delete a request, and only while Pending	New Request screen, re-checked right before saving
Rejecting needs at least one reason and a Pending request	Request Detail

9. Design system

Element	Value	Use
Primary green	#2E7D32	Title bars, main buttons, links, logo
Pending	#E68200	Status badge text
Done	#1E8C41	Status badge text
Rejected / danger	#C62828	Rejected badge, Reject and Delete buttons
Background	#F4F6F4	Screen background behind the white cards
Text / secondary text	#1C211D / #6B746C	Body text, labels

Components: full-width green title bar, white cards (12dp corners, light shadow), boxed input fields, rounded main button (green fill) and secondary button (green outline), green pill-shaped Back button at the bottom-left, rounded status badges.

10. Build and run

Item	Version
Android Studio	Current stable release (Android Gradle Plugin 8.5.0, Gradle 8.7)
Java	17 or newer (Android Studio's bundled JDK works)
Android SDK	compileSdk 34, targetSdk 34, minSdk 24 (Android 7.0)
Libraries	AppCompat 1.7.0, Material 1.12.0, ConstraintLayout 2.1.4, RecyclerView 1.3.2, CardView 1.0.0, Room 2.6.1

Open the project folder in Android Studio, wait for Gradle sync, choose an emulator and press Run. Details are in the Getting Started guide.

11. Manual test plan

There are no automated tests. Run these on the demo data; tick the box when a case passes.

ID	Steps	Expected result	Pass
T1	Login as tenant1 / tenant123	Tenant home opens with “Welcome back, Tenant 1”	☐
T2	Login with a wrong password	Toast: Username or password is incorrect	☐
T3	Login with empty fields	Toast: Please fill in all fields	☐
T4	Sign up a tenant: all fields, password abcd123, landlord username landlord1	Account created, returns to Login, new tenant can log in	☐
T5	Sign up with password abc12345 or abcdefgh	Toast: Password must have at least 4 letters and 3 numbers	☐
T6	Sign up with an existing username	Toast: Username already taken	☐
T7	Sign up as tenant with landlord username that does not exist	Toast: Landlord not found	☐
T8	Tenant submits a new request	Toast: Request submitted; request is Pending in the list	☐
T9	Tenant taps a Pending request, edits the title, saves	Toast: Request updated; list shows the new title	☐
T10	Tenant deletes a Pending request and confirms	Request disappears; Cancel keeps it	☐
T11	Tenant taps a Done request	Toast: Completed requests can't be edited	☐
T12	Login as landlord1, then landlord2	landlord1 sees Tenant 1 and 2 requests; landlord2 sees only Tenant 3's	☐
T13	Landlord opens a request and taps +	Tenant name, unit and email appear; − hides them	☐
T14	Landlord marks a Pending request Done	Badge turns green; button reads Mark as Pending	☐
T15	Landlord taps Reject Request, ticks nothing, presses Reject	Toast: Select at least one reason; dialog stays open	☐
T16	Landlord rejects with two reasons	Badge turns red; reasons card lists both	☐
T17	Tenant taps the rejected request	Dialog lists the reasons	☐
T18	Landlord taps Reopen Request	Back to Pending; reasons cleared	☐
T19	Keep landlord screen open, submit a request as a tenant	Notification appears and the list refreshes	☐
T20	Landlord taps Contact Tenant	Email app opens to the tenant's address with the title as subject	☐
T21	Scroll the lists with many requests	The whole page scrolls; buttons stay at the bottom	☐

12. Known limitations and future work

- Passwords are stored as plain text; they should be hashed.
- The new-request notification is shown to whichever landlord is logged in on the device, not only the tenant's landlord (the app has no server).
- No automated tests.
- Phone numbers are not collected, so the landlord can only contact a tenant by email.
- Requests cannot have photos.

13. Team

Person	Screen	Files
1	Login	LoginActivity, activity_login.xml, User, UserDao
2	Sign Up	SignupActivity, activity_signup.xml, AppDatabase, build.gradle
3	Tenant Home	TenantHomeActivity, activity_tenant_home.xml, RequestAdapter, item_request.xml
4	New Request	NewRequestActivity, activity_new_request.xml, Request, RequestDao
5	Landlord Home	LandlordHomeActivity, activity_landlord_home.xml, NewRequestReceiver
6	Request Detail	RequestDetailActivity, activity_request_detail.xml, AndroidManifest.xml, testing